

Q: Given input { 4371, 1323, 6173, 4199, 4344, 9679, 1989 } and a hash function $h(x) = x \pmod{10}$, Show Resulting

- Separate chaining hash table
- open addressing hash table using linear hashing
- open addressing hash table using quadratic hashing
- open addressing hash table using double hashing

Sol

a) SEPARATE CHAINING:

$$h(x) = x \pmod{10}$$

$$\bullet h(4371) = 4371 \% 10 = 1$$

$$\bullet h(1323) = 1323 \% 10 = 3$$

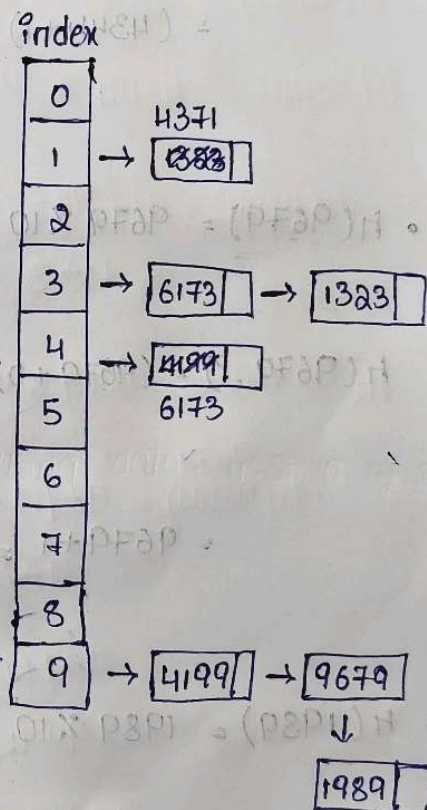
$$\bullet h(6173) = 6173 \% 10 = 3$$

$$\bullet h(4199) = 4199 \% 10 = 9$$

$$\bullet h(4344) = 4344 \% 10 = 4$$

$$\bullet h(9679) = 9679 \% 10 = 9$$

$$\bullet h(1989) = 1989 \% 10 = 9$$



b) LINEAR HASHING:

$$h(x) = h(\text{key} + i)$$

$$\bullet h(4371) = 4371 \% 10 = 1$$

$$\bullet h(1323) = 1323 \% 10 = 3$$

• $h(6173) = 6173 \% 10 = 3$
collision

$h(6173, i) = (6173 + 0) = 6173 \% 10 = 3$

$\Rightarrow p = 1$
collision

$= (6173 + 1) = 6174 \% 10 = 4$

$\Rightarrow p = 2$

• $h(4199) = 4199 \% 10 = 9$

• $h(4344) = 4344 \% 10 = 4$
collision

$h(4344, i) = (4344 + 0) = 4344 \% 10 = 4$

$\Rightarrow p = 1$ collision

$= (4344 + 1) = 4345 \% 10 = 5$

$\Rightarrow p = 2$

• $h(9679) = 9679 \% 10 = 9$

collision

$h(9679, i) = (9679 + 0) = 9679 \% 10 = 9$

$\Rightarrow p = 1$ collision

$= 9679 + 1 = 9680 \% 10 = 0$

$\Rightarrow p = 2$

• $h(1989) = 1989 \% 10 = 9$
collision

$h(1989, i) = (1989 + 0) = 1989 \% 10 = 9$

$\Rightarrow p = 1$ collision

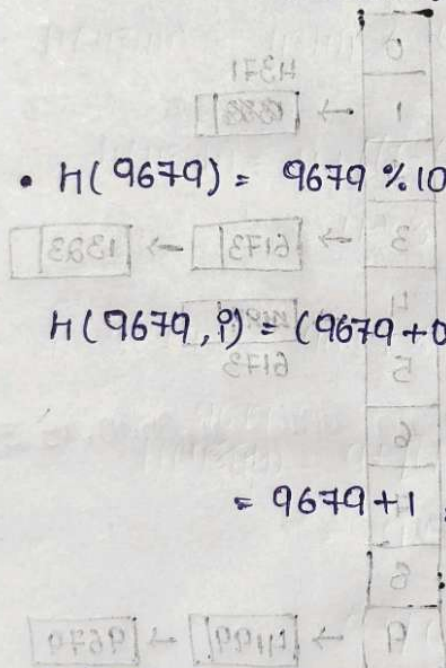
$= (1989 + 1) = 1990 \% 10 = 0$

$\Rightarrow p = 2$ collision

$= (1989 + 2) = 1991 \% 10 = 1$

$\Rightarrow p = 3$ collision

0	→ 9679
1	→ 4341
2	→ 1989
3	→ 1323
4	→ 6173
5	→ 4344
6	
7	
8	
9	→ 4199



$$\Rightarrow p = 40$$

c) QUADRATIC HASHING

$$H(\text{key}) = H(\text{key} + p)$$

$\hookrightarrow$ 1 to table size.

$$\bullet H(4371) = 4371 \% 10 = 1 //$$

$$\bullet H(1323) = 1323 \% 10 = 3 //$$

$$\bullet H(6173) = 6173 \% 10 = 3$$

collision

$$H(6173, i) = (6173 + i) = 6174 \% 10 = 4 //$$

$\Rightarrow p = 1$

$$\bullet H(4199) = 4199 \% 10 = 9 //$$

$$\bullet H(4344) = 4344 \% 10 = 4$$

collision

$$H(4344, i) = (4344 + i) = 4345 \% 10 = 5 //$$

$\Rightarrow p = 1$

$$\bullet H(9679) = 9679 \% 10 = 9$$

collision

$$H(9679, i) = (9679 + i) = 9680 \% 10 = 0 //$$

$\Rightarrow p = 1$

$$\bullet H(1989) = 1989 \% 10 = 9$$

collision

$$H(9679, i) = (1989 + i) = 1990 \% 10 = 0$$

$\Rightarrow p = 1$ collision

$$H(1989 + 4) = 1993 \% 10 = 3$$

$$\Rightarrow p = 2 \text{ collision}$$

$$H(1989 + 9) = 1998 \% 10 = 8 //$$

$$\Rightarrow p = 3$$

0	$\rightarrow$ 9679
1	$\rightarrow$ 4371
2	
3	$\rightarrow$ 1323
4	$\rightarrow$ 6173
5	$\rightarrow$ 4344
6	
7	
8	$\rightarrow$ 1989
9	$\rightarrow$ 4199

d) DOUBLE HASHING:

$$h(\text{key}) = [h_1(\text{key}) + (h_2(\text{key}) * i)] \quad \therefore R = \text{prime Number}$$

$$h_1(\text{key}) = x \pmod{t_s} \quad h_2(\text{key}) = R - x \pmod{R}$$

$$\bullet h(4371) = 4371 \% 10 = 1 //$$

$$R = 10 \Rightarrow 2, 3, 5, 7$$

$$\bullet h(1323) = 1323 \% 10 = 3 //$$

$$R = 7$$

$$\bullet h(6173) = 6173 \% 10 = 3$$

collision.

$$h(6173) = 3 + [(7-6) * i]$$

$$= 3 + 1 = 4 //$$

$$\bullet h(4199) = 4199 \% 10 = 9 //$$

$$\bullet h(4344) = 4344 \% 10 = 4$$

collision

$$h(4344) = 4 + [(7-4) * i]$$

$$= 4 + 3 = 7 //$$

$$\bullet h(9679) = 9679 \% 10 = 9$$

collision

$$h(9679) = 9 + [(7-5) * i]$$

$$= 9 + 2 = 11$$

collision

$$= 9 + [(7-5) * 2]$$

$$= 9 + 4 = 13$$

collision

$$= 9 + [(7-5) * 3]$$

$$= 9 + 6 = 15 = 5 //$$

$$\bullet h(1989) = 1989 \% 10 = 9$$

collision

$$h(1989) = 9 + [(7-1) * i]$$

$$= 9 + 6 = 15$$

collision

0	
1	→ 4371
2	
3	→ 1323
4	→ 6173
5	→ 9679
6	
7	→ 4344
8	
9	→ 4199

$$\begin{array}{r} 6173 \div 21 \\ 56 \\ \underline{57} \\ 86 \\ \underline{18} \\ 7 \\ \underline{6} \end{array}$$

$$= 9 + [(7-1) * 2]$$

$$= 9 + 12 = 21 \text{ collision}$$

$$= 9 + [(7-1) * 3]$$

$$= 9 + 18 = 27 \text{ collision.}$$

∴ 1989 does not have index position because i value has extended the tabular value.

Q09
Ans
12/8/20